Nonparametric Variational Regularisation of Pretrained Transformers

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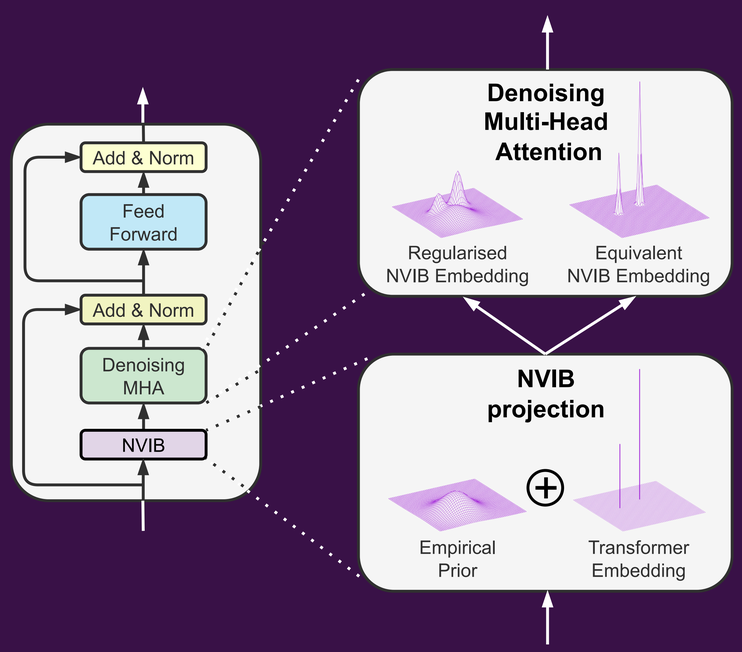
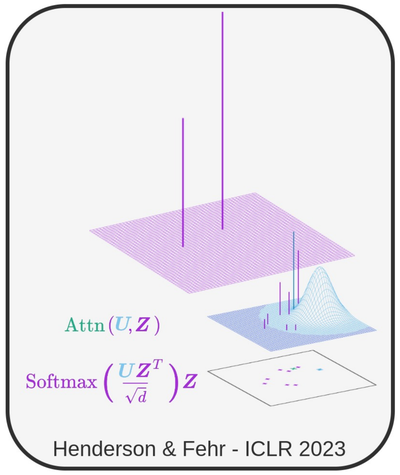
Use information theory to reinterpret **Transformers** and improve performance **without training**!

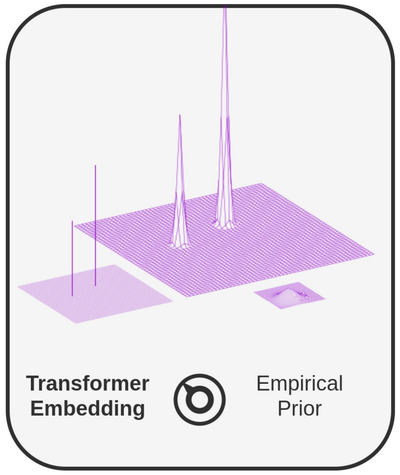


Background
NVIB



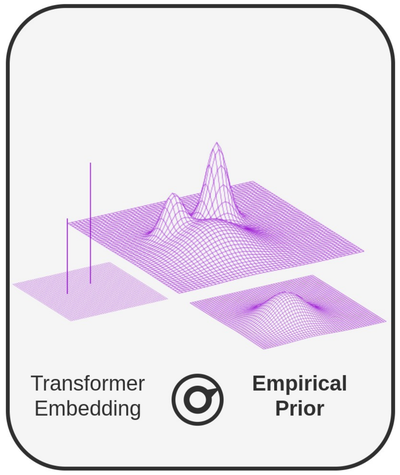
Nonparametric Variational Information Bottleneck (NVIB) formalises attention-based representions as mixture distributions and provides a VIB regulariser for Transformer embeddings.

Reinterpret
existing models



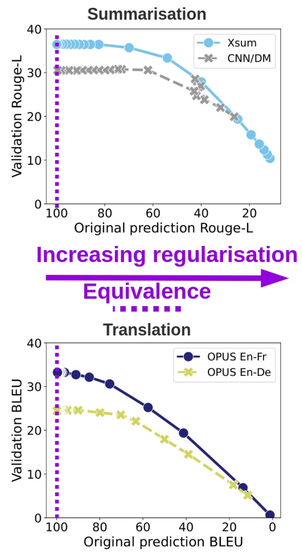
We **extend NVIB** to all forms of attention in Transformers.
We define an **empirical prior** over model embeddings for NVIB.
We reinterpret pretrained models using an **identity initialisation.**

Regularise
post-training



We define **controllable hyperparameters** to smoothly regularise.
Regularisation is done **post-training** without backpropagation.
We get **improved performance** on out-of-domain summarisation.

Smooth regularisation



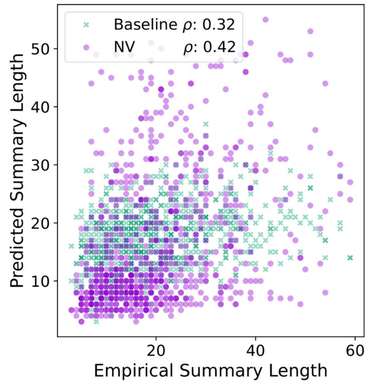
Improved out-of-domain generalization

Summarisation	↑%
NV-BART$_{(\mathrm{Xsum})}$	**+18.2 %**
NV-BART$_{(\mathrm{CNN/DM})}$	+2.3 %

Translation	↑%
NV-Marian$_{\mathrm{En\text{-}Fr}}$	+1.0 %
NV-Marian$_{\mathrm{En\text{-}De}}$	+0.4 %

Relative improvement over baseline

Adaptive generation



Model BART (Xsum)
Data SAMsum (Validation)

Paper

Code

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